

Seating & Event Availability Tracking System (SEATS) for La Loy Restaurant

Milestone 2 · Part 2 Final Deliverables: project scope statement, WBS and tasks, effort estimation, resource assignment and rates, cost estimate, schedule with Gantt chart and critical path

2110722 Software Project Management

Submitted to

Assoc. Prof. Twittie Senivongse

Members

SE69 Group 1 (regular program)

Boonnisa Tangjitpermkwamdee	Student ID 6970156821
Sirapatr Chalermwattanatrai	Student ID 6970292421
Chawanakorn Sanannam	Student ID 6970064721
Watayut Aiamanan	Student ID 6970267821

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Software Engineering Program, Department of Computer Engineering, Faculty of Engineering, Chulalongkorn University · Semester 1, Academic Year 2569

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Executive Summary

The Seating & Event Availability Tracking System (SEATS, called CTRS in Milestone 1) is the software house's LINE-integrated reservation and check-in system, built first for La Loy Restaurant under a pilot implementation agreement and then productized as a subscription service. Milestone 1 set the business case, the charter (3 August 2026 to 28 July 2028, budget 1,010,590 THB) and a top-down estimate of 2,632 engineer hours. This Milestone 2 report turns that charter into the planning baselines of the project: the scope baseline (scope statement, WBS and WBS dictionary), the effort and resource plan, the cost baseline and the schedule baseline.

- **Scope.** Three releases, 1.0 Core (features F1 to F28), 2.0 Growth (F29 to F35) and 2.1 Product (multi-tenant productization), with the exclusions F36, F37 and configurable business rules, the deliverables of the tailored RUP and the seven success criteria of the charter.
- **WBS and effort.** 49 work packages in eight branches, each with one accountable engineer, bottom-up hours per engineer and three-point estimates for the four risky packages; the bottom-up total is the 2,632 hours of Milestone 1, re-allocated within its phase totals so that the budget of the charter holds.
- **Resources and cost.** The four part-time engineers at 300 and 350 THB an hour, the test devices, hosting, domain and legal set-up; labor 846,600 THB, other costs 75,500 THB, contingency reserve 88,490 THB tied to the risks of Milestone 1; cost baseline and project budget 1,010,590 THB, no management reserve; time-phased by iteration.
- **Schedule.** 49 activities in the ten iterations plus the pilot year and close-out, with predecessors, resource units and seven milestones; the critical path runs through every iteration up to Release 2.0 and then through the pilot year to the expansion decision on 28 July 2028; the productization branch (C4, T3) has 115 working days of total slack.

Document Revision History

Version	Date	Description	Author
1.0	10 September 2026	Milestone 1: Review Board Decision Log, business case, financial analysis, project charter, project organization chart (project then named CTRS)	SE69 Group 1
2.0	15 October 2026	Milestone 2: Review Board Decision Log, project scope statement, WBS and tasks, effort estimation, resource assignment and rates, cost estimate, schedule with Gantt chart and critical path; project renamed SEATS	SE69 Group 1

1. Review Board Decision Log

The four members hold one Review Board meeting per artifact set, after each member's prompting session for the Milestone 2 deliverables. Every artifact is reviewed against the three criteria of the brief, Requirement Satisfaction, PMBOK Correctness and Project Consistency (with the Milestone 1 report and with the other Milestone 2 artifacts), and the board keeps the role separation of the tailored RUP: each artifact has exactly one accountable owner and the reviewer is never the author. Table 1.1 records one row per artifact or artifact part with the board's suggestion and decision; the prompt design, review and revision of each artifact are documented in the GenAI Log file (Part 1), and every prompt draws on the same single source of truth, the Milestone 1 report (dates, release scope, rates, phase effort and budget).

Table 1.1 Review Board Decision Log, Milestone 2

No.	Artifact	Member (prompt design)	Suggestion from team (in brief)	Decision (Accept / Revise / Reject)
1	Scope statement and deliverables (Section 2)	Boonnisa T.	The draft mixed project-management, product and closure deliverables and did not list exclusions. Separate the deliverable categories, state the exclusions and retain the release boundaries.	Revise
2	Success criteria and budget baseline (Sections 2.5 and 6)	Boonnisa T.	Success measures needed numeric targets and evidence. Use the source targets (booking within 3 minutes, no-show reduction of 35 %, at least 5 transactions a second, response within 2 seconds, check-in within 15 seconds per table, recovery within 2 hours) and keep the budget at 1,010,590 THB.	Accept
3	WBS and schedule structure (Sections 3 and 7)	Sirapatr C.	The first WBS was phase-based with verb-action labels and no schedule columns. Re-decompose the workstreams to two levels with deliverable-oriented noun phrases and add WBS ID, task, responsible role, duration, predecessors, start, finish and expected output, preserving the iteration dates of Milestone 1.	Revise

No.	Artifact	Member (prompt design)	Suggestion from team (in brief)	Decision (Accept / Revise / Reject)
4	Milestones and critical path (Sections 7.2 and 7.3)	Sirapatr C.	Milestone 1 gives iteration dates but no formally calculated critical path. Derive the chain with the standard float formulas (TS = LS − ES = LF − EF), show the zero-slack path, keep the productization branch as parallel work with float, and qualify the result as a planning analysis.	Accept
5	Effort estimation for tasks (Section 4)	Chawanakorn S.	The C1 payment-gateway effort was too low against the Milestone 1 baseline and the basis of estimates was missing. Reconcile the work-package allocations to 2,632 hours and 846,600 THB (0 % variance), state the basis of estimates and use three-point / PERT for the gateway, LINE and load-test items.	Revise
6	WBS decomposition and features by release (Section 3)	Chawanakorn S.	Core, Growth and productization features were mixed and some items were listed by page. Cover every feature F1 to F35 and the C4 / T3 productization exactly once, use noun-form work packages and one numbering scheme across all tables.	Revise
7	Quality requirements and tests (Sections 2.2 and 2.5)	Watayut A.	Quality requirements lacked verification methods. Map response time, throughput, check-in, recovery, backup and browser compatibility to test evidence with measurable thresholds.	Revise
8	Cost estimate consistency (Section 6)	Watayut A.	The budget needed transparent categories, exclusions and a final arithmetic check. Preserve the reported figures (Year 0 767,690; Year 1 242,900; total 1,010,590; labor 846,600; pilot fee 150,000; net 860,590) and flag the reconciliation for team verification.	Accept

Artifacts marked Revise were revised as suggested before submission, and the sections of this document are the revised versions; every revision is recorded in the Review and Revision fields of the artifact's GenAI Log record (Part 1), and every artifact was accepted by the board into the project baseline after its revision.

2. Project Scope Statement

2.1 Project Summary

SEATS is a LINE-integrated web application for reserving tables at La Loy Restaurant's live-music concerts. Customers use a LIFF web application opened from the Rich Menu of the restaurant's LINE Official Account to browse concert rounds, view a live table map, select a table, hold it for 15 minutes, pay the full table fee through a payment gateway and receive a QR e-ticket. Restaurant staff use a shared back-office to manage rounds, layouts and prices, watch the live booking view, handle exceptions and scan QR codes at the door. One central record replaces the chat, the hand-edited image and the notebook, so that double bookings and payment disputes disappear and the Manager handles exceptions only. After the build, the system is run through a pilot year at La Loy and productized as a multi-tenant subscription service for other live-music restaurants.

The project runs from 3 August 2026 to 28 July 2028 (two project years) under the tailored Rational Unified Process of the Development Case: Year 0 builds and deploys Release 1.0 Core (in operation 2 April 2027) and Release 2.0 Growth (30 July 2027); Year 1 runs the live pilot and delivers Release 2.1 Product (11 February 2028), and closes with the expansion decision of the software house on 28 July 2028. The project budget is 1,010,590 THB (charter, Table 4.1 of Milestone 1), of which La Loy's pilot implementation fee covers 150,000 THB. The scope statement, the WBS of Section 3 and the WBS dictionary of Section 3.3 form the scope baseline; changes to it go through the Change Control Board.

2.2 Product Characteristics and Requirements

The product scope is the feature set of the Vision (Tables 2.15 to 2.18 of Milestone 1), grouped by release and user role in Table 2.1; the detailed requirements are the use-case specifications, the Software Requirements Specification and the Supplementary Specification of the Requirements Engineering project. The quality characteristics (Table 2.2) are the preliminary quality targets of Milestone 1 (Table 2.19), each now tied to the test that verifies it, as the Review Board asked (Table 1.1, row 7). Personal data is handled under the Personal Data Protection Act, with consent, retention and correction designed in; drink packages are presented within the Alcoholic Beverage Control Act.

Table 2.1 Product scope by release and role

Release	Role	Features	Characteristics
1.0 Core	Customer	F1 LINE sign-in, F2 table map, F3 table selection and 15-minute hold, F4 gateway payment, F5 QR e-ticket, F6 transactional messages, F7 booking terms, F8 my bookings, F9 round list	Self-service booking inside LINE: browse, choose, hold, pay the full fee, keep the e-ticket; no Manager reply needed
1.0 Core	Manager (Owner read-only)	F10 concert rounds, F11 zones and packages, F12 live booking view, F13 ticket escalation management, F14 manual booking and payment overrides	Set up each concert round and watch every table, hold, payment and check-in on one live screen; exceptions by hand, including the degraded payment mode
1.0 Core	Front Staff	F15 QR scan, F16 ticket details, F17 on-site check-in, F18 invalid-ticket escalation, F19 no-show and table release, F20 name lookup	Check-in by QR within 15 seconds per table; release of no-show tables after the grace period
1.0 Core	System	F21 hold release, F22 daily backup, F23 payment callback, F24 staff accounts, F25 audit trail, F26 push-message budget, F27 degraded-mode switch, F28 consent and profile access	First-lock-wins hold with timeout, automatic confirmation on the gateway's webhook, role-based access, audit trail, backup with Excel export, consent
2.0 Growth	Customer and System	F29 grace extension, F30 waitlist, F31 booking transfer, F32 scheduled reminders, F34 reminder scheduling, F35 waitlist offer	Use the central record to lose fewer tables: reminders, grace extension, waitlist in queue order, official transfer
2.0 Growth	Manager and Owner	F33 reports	Booking, revenue, artist-popularity and customer-history reports with export
2.1 Product	Software house	Multi-tenant data model and tenant isolation; Free, Standard and Premium plans with limits and enforcement; subscription billing and setup-fee invoicing; self-service onboarding; service administration; migration of La Loy	The pilot system productized as a subscription service within the 1,000 THB a month ceiling (Section 3.1 of Milestone 1)

Table 2.2 Quality requirements and their verification

Quality requirement	Target (Table 2.19 of Milestone 1)	Verification (test evidence)	When
Throughput at booking-open time	at least 5 transactions a second	Load test with simulated customers opening a round: 5 transactions a second sustained for 10 minutes, one hold per table, no deadlock (WBS 4.10)	before IOC; repeated for Release 2.1 (8.5)

Quality requirement	Target (Table 2.19 of Milestone 1)	Verification (test evidence)	When
Response time	within 2 seconds for customer and Front Staff transactions	Response-time measurement during the load test and in the acceptance tests on the test devices inside LINE (4.10, 5.1)	IOC, T1
Check-in time	within 15 seconds per table	Timed check-in of 30 tickets by the Front Staff with the tablet at a pilot night (5.1, 5.4)	T1
Recovery after an outage	within 2 hours during opening hours	Restore drill: restore the daily backup to a fresh environment and bring the service up within 2 hours (4.10, repeated in 8.5)	IOC, C4
Backup and export	daily backup of booking data with export to Excel	Backup job verified daily in the test environment; the Excel export opened by the Manager (4.9, 5.1)	C2, T1
Supported browsers	current LINE in-app browser; last two versions of Chrome, Safari and Edge	Browser-compatibility checklist run on the test devices and the shop's desktop browser (4.10, 5.1)	IOC, T1
Operating cost of the restaurant	hosting, APIs and LINE plan at or below 1,000 THB a month	Monthly cost review of hosting and push messages during the pilot year (1.8)	P1

2.3 Project Deliverables

Table 2.3 Project deliverables

Category	Deliverables
Project-management deliverables	Business case with financial analysis, project charter, stakeholder list and project organization (Milestone 1); scope statement, WBS and WBS dictionary, effort, resource and cost estimates, schedule and cost baseline (this document); quality, risk and communication management plans (Milestone 3); Development Case and Software Development Plan; iteration plan and assessment of every iteration; weekly status assessments; change-request decisions of the Change Control Board; Risk List updates; milestone reviews (LCO, LCA, IOC, Product Release 1.0, 2.0 and 2.1); monthly pilot KPI reports; pilot review, expansion decision record, hand-over record, lessons learned and sign-off

Category	Deliverables
Product-related deliverables	Requirements and design artifacts of the tailored RUP: Vision, Glossary, Stakeholder Requests, Business Rules, Domain Model, Use-Case Model and use-case specifications, Software Requirements Specification, Supplementary Specification, Analysis Model, Software Architecture Document, user-interface prototype, test plan and test cases; the tested increment of every Construction iteration; Release 1.0 Core in operation at La Loy (2 April 2027), Release 2.0 Growth (30 July 2027) and the productized Release 2.1 (11 February 2028); load-test, system-test and acceptance records; training and user guide for the Manager and the Front Staff; migration of the current bookings and, in Release 2.1, of La Loy to the productized service; production environment, support routine and Free-plan sign-up

2.4 Exclusions

- Split payment among the members of a party (F36) and table photos with distance to the stage (F37): Future candidates outside the project (Table 2.13 of Milestone 1).
- Restaurant-configurable booking rules beyond the rules agreed for the defined releases (hold, grace, transfer cut-off, waitlist, refund policy are fixed by the Owner's decisions before LCA).
- Payment-gateway fees, each restaurant's LINE Official Account plan and the pilot restaurant's own check-in tablet (about 9,000 THB): paid by the restaurant (CON-03, CON-07), not by the project budget.
- Sales, onboarding and operation of the subscription service for other restaurants after the expansion decision (Years 2 to 5 of the financial analysis): outside the project.
- Hardware at the venue beyond the test devices of the software house; any change to the restaurant's own LINE Official Account content.

2.5 Project Success Criteria

The success criteria are the seven criteria of the charter (Table 4.1 of Milestone 1), each with the evidence that shows it is met (Table 2.4). They are also the acceptance criteria of the major deliverables: Release 1.0 is accepted when criteria 2, 3, 5 and 6 are shown on the pilot nights of T1; Release 2.0 when the acceptance tests of F29 to F35 pass; Release 2.1 when La Loy runs on the productized service and the per-shop cost assumption holds.

Table 2.4 Success criteria and evidence

No.	Success criterion (charter)	Target / evidence
1	Release milestones within budget	Release 1.0 Core in operation at La Loy on 2 April 2027, Release 2.0 Growth on 30 July 2027, Release 2.1 on 11 February 2028; cost at close-out within the project budget of 1,010,590 THB (go-live records, iteration assessments, cost baseline of Section 6)
2	Booking integrity	Zero double bookings on La Loy's concert nights after go-live; table status real-time on one record (audit trail and pilot KPI reports)
3	Customer experience	A booking completed within 3 minutes from table selection to confirmation, without a Manager reply (timestamps of the booking record; sample of pilot nights)
4	Business outcome	No-show rate at La Loy reduced by 35 % within six months of Release 2.0, against the baseline of Table 2.2 of Milestone 1 (check-in records versus bookings)
5	Operational quality and cost	Check-in within 15 seconds per table; at least 5 transactions a second at booking-open time; response within 2 seconds; recovery within 2 hours; the restaurant's hosting, APIs and LINE plan at or below 1,000 THB a month (Table 2.2 tests; monthly cost review)
6	Requirements and acceptance	Vision approved by the Owner at LCO, SRS signed off at LCA, acceptance tests of the two main use cases (reserve a concert table, check in with e-ticket) passed with the Manager and the Front Staff (milestone review and acceptance records)
7	Expansion gate	At the expansion decision: La Loy continues on a paid plan, at least 10 restaurants are on the waiting list, and the per-shop hosting and support cost of the productized service is at or below the assumptions of Table 3.3 of Milestone 1 (pilot review, waiting list, cost report)

3. WBS and Tasks

3.1 Work Breakdown Structure and Activity List

The WBS (Table 3.1) is a deliverable-oriented decomposition of the total scope: level 1 is the project, level 2 the eight branches (project management, the phases of the tailored RUP by release, the pilot operation), level 3 the 49 work packages, named as nouns, each carried out by one activity (verb phrase). It follows the rules of the course: every unit of work appears in exactly one work package (100 % rule, checked for F1 to F35 and the Release 2.1 items), the work of a branch is the sum of its work packages, each work package has one accountable engineer (Table 4.1) and a WBS dictionary entry (Section 3.2), and the structure matches how the work is performed, by iteration. Project management is one branch that runs through all phases, so that status assessments, iteration assessments, change control, milestone reviews, pilot KPI reports and close-out are visible as work of their own. The pilot support of the restaurant (7.1) is work of the team but operating expenditure of Year 1 (Table 3.6 of Milestone 1), not development effort. Each activity has a duration in working days, a start and finish inside the iteration dates of Milestone 1 (Table 2.21 there), its predecessors (finish-to-start unless marked SS, start-to-start, or FF, finish-to-finish) and the engineers assigned with their units (PM Boonnisa T., SA Sirapatr C., RS Chawanakorn S., QA Watayut A.; 100 % is 8 hours a day); the seven milestones have zero duration and the dates of the charter. The calendar is Monday to Friday with the year-end break of 21 December 2026 to 1 January 2027; Thai public holidays are absorbed by the part-time allocation. Work packages are planned in detail up to Release 2.0; C4, T3 and the pilot year are planned at work-package level and will be detailed before C4 (rolling wave planning).

Table 3.1 Work breakdown structure and activity list of the SEATS project

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
1	Project Management						
1.1	**Project plans and baselines** (I1, I2)	Produce the business case, the project charter, the organization, the Software Development Plan and the iteration plans I1 and I2	23	3 Aug 2026	2 Sep 2026	—	PM 11 %
1.2	**Risk List and Development Case** (I1, I2)	Write the initial Risk List (RISK-01 to RISK-14) and the Development Case that tailors RUP for the project	18	10 Aug 2026	2 Sep 2026	2.1 SS	PM 3 %, QA 6 %

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
1.3	**Iteration assessments and change control, Elaboration** (E1, E2)	Run the weekly status assessments, the E1 and E2 iteration assessments and the Change Control Board	37	3 Sep 2026	23 Oct 2026	1.1	PM 9 %
1.4	**Iteration assessments and change control, Release 1.0 Construction** (C1, C2)	Run the weekly status assessments, the C1 and C2 iteration assessments and the Change Control Board	80	26 Oct 2026	26 Feb 2027	1.3	PM 10 %
1.5	**Iteration assessments and change control, Release 1.0 Transition** (T1)	Run the weekly status assessments, the T1 assessment and the Change Control Board	25	1 Mar 2027	2 Apr 2027	1.4	PM 6 %
1.6	**Iteration assessments and change control, Release 2.0** (C3, T2)	Run the weekly status assessments, the C3 and T2 iteration assessments and the Change Control Board	85	5 Apr 2027	30 Jul 2027	1.5	PM 7 %
1.7	**Iteration assessments and change control, Release 2.1** (C4, T3)	Run the status assessments, the C4 and T3 iteration assessments and the Change Control Board	140	2 Aug 2027	11 Feb 2028	1.6	PM 2 %
1.8	**Pilot KPI reports and reviews** (P1)	Collect the pilot KPIs of every concert night (booking time, double bookings, no-shows, check-in time, operating cost) and review them monthly with the Owner	255	2 Aug 2027	21 Jul 2028	6.7	PM 1 %, QA 0 %
1.9	**Pilot review, expansion decision and close-out** (CL)	Review the pilot year against the success criteria of the charter, prepare the expansion decision, record lessons learned and obtain sign-off	5	24 Jul 2028	28 Jul 2028	1.8 FF, 8.6	PM 20 %, SA 10 %, QA 10 %
2	**Inception (I1, I2)**						
2.1	**Stakeholder requests and problem analysis** (I1)	Interview the Owner and the Manager, observe concert nights, record the stakeholder requests, the problems and their root causes (5 Whys)	15	3 Aug 2026	21 Aug 2026	—	PM 3 %, SA 13 %, RS 7 %
2.2	**Vision and Glossary** (I1, I2)	Write the Vision (problem statement, stakeholders, features F1 to F37 with value, cost and priority) and the Glossary	18	10 Aug 2026	2 Sep 2026	2.1 SS	SA 14 %, RS 8 %
2.3	**Use-case model and release plan** (I2)	Outline the use-case model and assign the features to Release 1.0, Release 2.0 and Future	7	25 Aug 2026	2 Sep 2026	2.2 SS	SA 7 %, RS 21 %, QA 7 %
2.4	**Review records and LCO package** (I2)	Review the Inception artifacts against the RUP checkpoints, record the Review Records and assemble the LCO review	5	27 Aug 2026	2 Sep 2026	2.2 SS, 2.3 SS	PM 10 %, QA 30 %
3	**Elaboration (E1, E2)**						

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
3.1	**Use-case specifications** (E1)	Specify the two main use cases, Reserve a concert table and Check in with e-ticket, with the Manager and the Front Staff	10	3 Sep 2026	16 Sep 2026	2.4	SA 20 %, RS 40 %, QA 5 %
3.2	**Software Requirements Specification and Supplementary Specification** (E1, E2)	Write the SRS and the Supplementary Specification with the quality targets of Table 2.19 of Milestone 1	22	10 Sep 2026	9 Oct 2026	3.1 SS	PM 2 %, SA 7 %, RS 16 %, QA 5 %
3.3	**Analysis model and Software Architecture Document** (E2)	Produce the analysis model and the Software Architecture Document (LIFF front end, back office, application server, central database, LINE and gateway integration)	22	17 Sep 2026	16 Oct 2026	3.1	PM 2 %, SA 27 %, RS 5 %
3.4	**User-interface prototype** (E2)	Prototype the customer and back-office screens and have the Manager, the Front Staff and sample customers try them	17	24 Sep 2026	16 Oct 2026	3.1	SA 3 %, RS 18 %, QA 3 %
3.5	**Payment gateway and LINE sandbox** (E2)	Open the gateway sandbox and the LINE Developers channel, verify the merchant onboarding and the LIFF and Messaging API settings (RISK-03, external dependency)	17	17 Sep 2026	9 Oct 2026	2.4	PM 6 %, SA 9 %, RS 3 %
3.6	**Test plan and test cases** (E2)	Write the test plan and the test cases of the two main use cases, including the load-test design for 5 transactions a second (RISK-05)	22	24 Sep 2026	23 Oct 2026	3.1	PM 2 %, QA 18 %
3.7	**Requirements reviews, SRS baseline and LCA package** (E2)	Review the SRS, the SAD, the prototype and the test plan against the checkpoints, baseline the SRS and assemble the LCA review	10	12 Oct 2026	23 Oct 2026	3.2, 3.3 SS, 3.4 SS, 3.6 SS	PM 10 %, SA 5 %, QA 20 %
3.8	**Scope baseline, estimates, schedule and cost baseline** (E2)	Produce the scope statement, the WBS and dictionary, the effort, resource and cost estimates, the schedule and cost baseline (this document) and the Construction plan	10	12 Oct 2026	23 Oct 2026	3.2 SS	PM 10 %
4	**Release 1.0 Core: Construction (C1, C2)**						
4.1	**Concert rounds, zones and packages** (C1)	Design, implement and unit-test the management of concert rounds, zones, table types, packages and prices	15	26 Oct 2026	13 Nov 2026	3.7	PM 7 %, SA 13 %, RS 27 %, QA 7 %

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
4.2	**Table map, selection and hold** (C1)	Implement the round list, the real-time table map, the table selection with the 15-minute hold and the automatic hold release	20	2 Nov 2026	27 Nov 2026	4.1 SS	PM 5 %, SA 20 %, RS 22 %, QA 10 %
4.3	**Gateway payment and callback** (C1)	Integrate the hosted checkout, the booking terms, the payment webhook and the automatic confirmation of the booking	25	16 Nov 2026	18 Dec 2026	3.5, 4.2 SS	PM 4 %, SA 20 %, RS 16 %, QA 12 %
4.4	**LINE sign-in, staff accounts and role access** (C1)	Implement LINE Login, the back-office accounts and the role-based access of Manager, Front Staff and Owner	15	16 Nov 2026	4 Dec 2026	3.7	SA 13 %, RS 13 %, QA 7 %
4.5	**C1 integration and iteration test** (C1)	Integrate the C1 increment, run the iteration tests and the C1 assessment	10	7 Dec 2026	18 Dec 2026	4.1, 4.2, 4.3 SS, 4.4	PM 10 %, SA 10 %, RS 10 %, QA 30 %
4.6	**QR e-ticket and transactional messages** (C2)	Implement the QR e-ticket, the customer's booking list and the transactional LINE messages	15	4 Jan 2027	22 Jan 2027	4.5	SA 13 %, RS 27 %, QA 7 %
4.7	**Door check-in and name lookup** (C2)	Implement the QR scan, the ticket details, the check-in, the invalid-ticket escalation, the no-show release and the name lookup	20	11 Jan 2027	5 Feb 2027	4.6 SS	PM 5 %, SA 10 %, RS 20 %, QA 10 %
4.8	**Live booking view and exception handling** (C2)	Implement the live booking view, the ticket escalation, the manual overrides and the degraded-mode switch	20	18 Jan 2027	12 Feb 2027	4.6	PM 5 %, SA 20 %, RS 20 %, QA 10 %
4.9	**Audit trail, backup, consent and push budget** (C2)	Implement the audit trail, the daily backup with Excel export, the consent and profile access and the push-message budget	15	25 Jan 2027	12 Feb 2027	4.6	PM 7 %, SA 13 %, RS 17 %, QA 7 %
4.10	**Load test, system test and IOC package** (C2)	Run the load test (5 transactions a second at booking-open time), the system test of Release 1.0 and assemble the IOC review	15	8 Feb 2027	26 Feb 2027	4.7, 4.8, 4.9	PM 7 %, SA 27 %, RS 13 %, QA 53 %
5	**Release 1.0 Core: Transition (T1)**						
5.1	**Acceptance tests of Release 1.0** (T1)	Run the acceptance tests of the two main use cases with the Manager and the Front Staff	10	1 Mar 2027	12 Mar 2027	4.10	PM 10 %, SA 10 %, RS 15 %, QA 35 %

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
5.2	**Training and user guide** (T1)	Train the Manager and the Front Staff and hand over the user guide	10	8 Mar 2027	19 Mar 2027	5.1 SS	PM 10 %, RS 20 %, QA 5 %
5.3	**Production set-up and data migration** (T1)	Set up the production hosting, the domain and the gateway merchant account of La Loy, and migrate the current bookings	10	15 Mar 2027	26 Mar 2027	5.1	SA 30 %, RS 12 %, QA 10 %
5.4	**Pilot concert nights and go-live of Release 1.0** (T1)	Support the first concert nights on site, fix the defects found and declare Release 1.0 in operation	10	22 Mar 2027	2 Apr 2027	5.2, 5.3	PM 15 %, SA 10 %, RS 15 %, QA 25 %
6	**Release 2.0 Growth (C3, T2)**						
6.1	**Scheduled reminders** (C3)	Implement the reminder scheduling and the scheduled LINE reminders within the push-message budget	25	5 Apr 2027	7 May 2027	5.4	PM 4 %, SA 16 %, RS 20 %, QA 10 %
6.2	**Grace extension and waitlist** (C3)	Implement the grace extension, the waitlist and the waitlist offer (rules decided by the Owner, RISK-08)	30	26 Apr 2027	4 Jun 2027	6.1 SS	PM 3 %, SA 17 %, RS 20 %, QA 12 %
6.3	**Booking transfer** (C3)	Implement the official transfer of a confirmed booking to another LINE account	15	17 May 2027	4 Jun 2027	6.1	SA 13 %, RS 20 %, QA 10 %
6.4	**Reports and Excel export** (C3)	Implement the booking, revenue, artist-popularity and customer-history reports with export to Excel	20	24 May 2027	18 Jun 2027	6.1	PM 5 %, SA 15 %, RS 20 %, QA 10 %
6.5	**C3 integration and system test** (C3)	Integrate Release 2.0, run the regression and system tests and the C3 assessment	15	7 Jun 2027	25 Jun 2027	6.2, 6.3, 6.4	PM 10 %, SA 27 %, RS 20 %, QA 37 %
6.6	**Acceptance tests and training on Release 2.0** (T2)	Run the acceptance tests of the new features with the Manager and train the staff on them	15	28 Jun 2027	16 Jul 2027	6.5	PM 7 %, SA 12 %, RS 20 %, QA 25 %
6.7	**Release 2.0 deployment and first concert nights** (T2)	Deploy Release 2.0, support the first concert nights and declare Release 2.0 in operation	15	12 Jul 2027	30 Jul 2027	6.6 SS	PM 10 %, SA 13 %, RS 13 %, QA 17 %
7	**Pilot operation (P1)**						

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
7.1	**Concert-night support and maintenance** (P1)	Support the pilot restaurant on concert nights (on-call engineer) and maintain the system: 8 hours a month, costed as operating expenditure of Year 1 in Milestone 1 (Table 3.6), outside the 2,632 development hours	260	2 Aug 2027	28 Jul 2028	6.7	RS 2 %, QA 2 %
8	**Release 2.1 Product (C4, T3)**						
8.1	**Multi-tenant data model and tenant isolation** (C4)	Design and implement the multi-tenant data model, the tenant isolation and the migration path of La Loy's data	40	2 Aug 2027	24 Sep 2027	6.7	SA 12 %, RS 12 %, QA 5 %
8.2	**Plans, limits and tier enforcement** (C4)	Implement the Free, Standard and Premium plans, their limits and the automatic enforcement of the limits	30	20 Sep 2027	29 Oct 2027	8.1	SA 8 %, RS 17 %, QA 8 %
8.3	**Subscription billing and company set-up** (C4)	Implement the subscription billing and the setup-fee invoicing; register the company and publish the terms of service, privacy policy and data-processing agreement	30	25 Oct 2027	3 Dec 2027	8.2 SS	PM 3 %, SA 10 %, RS 17 %, QA 10 %
8.4	**Self-service onboarding and service administration** (C4)	Implement the self-service onboarding of a restaurant (zone map, packages, gateway and LINE set-up) and the service administration console	30	15 Nov 2027	24 Dec 2027	8.2	SA 8 %, RS 17 %, QA 8 %
8.5	**Security, load and integration test of Release 2.1** (C4)	Run the multi-tenant security test, the load test and the system test of Release 2.1 and the C4 assessment	25	13 Dec 2027	14 Jan 2028	8.3, 8.4 SS	PM 4 %, SA 6 %, RS 10 %, QA 14 %
8.6	**Migration of La Loy and release of the Free-plan sign-up** (T3)	Migrate La Loy to the productized service, release the Free-plan sign-up and hand over to the support routine	20	17 Jan 2028	11 Feb 2028	8.5	PM 2 %, SA 8 %, RS 12 %, QA 12 %
M1	*LCO: Lifecycle Objectives*		0	2 Sep 2026	2 Sep 2026	2.4	—
M2	*LCA: Lifecycle Architecture, SRS baseline*		0	23 Oct 2026	23 Oct 2026	3.7	—
M3	*IOC: Initial Operational Capability*		0	26 Feb 2027	26 Feb 2027	4.10	—
M4	*Release 1.0 Core in operation*		0	2 Apr 2027	2 Apr 2027	5.4	—
M5	*Release 2.0 Growth in operation*		0	30 Jul 2027	30 Jul 2027	6.7	—

WBS ID	Work package (iteration)	Activity	Days	Start	Finish	Predecessors	Resources (units)
M6	*Release 2.1 in operation*		0	11 Feb 2028	11 Feb 2028	8.6	—
M7	*Expansion decision, project closed*		0	28 Jul 2028	28 Jul 2028	1.9	—

3.2 WBS Dictionary

Every work package has a WBS dictionary entry with the fields of Table 3.2, kept in the project's planning file; the entry printed is the work package the Review Board found most critical, the payment gateway integration (RISK-03, RISK-06).

Table 3.2 WBS dictionary entry of work package 4.3

Field	WBS 4.3 Gateway payment and callback
WBS item number	4.3
WBS item name	Gateway payment and callback
Description	Integrate the hosted checkout, the booking terms, the payment webhook and the automatic confirmation of the booking.
Responsibility	Sirapatr Chalermwattanatrai (System Analyst, Software Architect), accountable; Boonnisa T., Chawanakorn S., Watayut A. contribute
Input	SRS and SAD baseline (3.7); sandbox and merchant account of 3.5; table hold of 4.2
Output	Payment flow with automatic confirmation
Features / content	F4, F7, F23
Effort	104 h (Boonnisa T. 8, Sirapatr C. 40, Chawanakorn S. 32, Watayut A. 24); method: three-point, O/M/P 80/100/150 h, PERT 105.0 h
Duration	25 working days, 16 Nov 2026 to 18 Dec 2026
Predecessors	3.5, 4.2 SS
Labor cost	33,600 THB
Acceptance criteria	A sandbox payment of the full table fee confirms the booking automatically within 10 seconds of the gateway's webhook; a timed-out hold rejects the payment; the webhook signature is verified; the degraded-mode switch (4.8) takes over when the gateway does not answer

4. Effort Estimation

4.1 Basis of Estimates

Effort is estimated in person-hours per work package, bottom-up, by the engineer who will do the work and reviewed by the Quality Assurance role; duration is derived from the effort and the engineer's part-time availability, not the other way round. The assumptions are those of Milestone 1: the four engineers work part-time at the hours a week per phase of Table 3.4 of Milestone 1 (for example 16 hours a week for the Requirements Specifier in Construction), at the software house's internal cost rates of 300 and 350 THB an hour; a working week has five days of 8 hours. Two techniques are used and compared, as the course asks: the bottom-up estimate per work package (Table 4.1, with three-point estimates for the four risky packages, Table 4.2) and the top-down estimate per phase of Milestone 1 (Table 4.3). The Review Board decided (Table 1.1, row 5) to allocate the bottom-up hours within the phase totals of Milestone 1, so that the labor cost stays at 846,600 THB and the cost baseline at the budget of the charter; the estimate therefore moves from the budgetary accuracy of Milestone 1 (−10 % to +25 %) toward a definitive estimate (−5 % to +10 %) for the iterations up to Release 2.0, while C4 and T3 remain at budgetary accuracy until they are detailed before C4.

4.2 Bottom-up Estimate per Work Package

Table 4.1 gives the accountable engineer, the hours of each engineer and the labor cost of every work package at the rates of Table 5.1 (the packages estimated three-point, Table 4.2, are starred); it is also the assignment of the human resources to the work packages (Section 5.2). The 96 hours of concert-night support (7.1) are shown for completeness but are operating expenditure (28,800 THB in Table 3.6 of Milestone 1), outside the 2,632 development hours.

Table 4.1 Effort and assignment per work package and engineer

WBS ID	Work package (* three-point, Table 4.2)	Accountable	Boonnisa T. (PM)	Sirapatr C. (SA)	Chawanakorn S. (RS)	Watayut A. (QA)	Hours	Labor (THB)
1	**Project Management**							
1.1	Project plans and baselines	Boonnisa T.	20	-	-	-	20	7,000
1.2	Risk List and Development Case	Watayut A.	4	-	-	8	12	3,800
1.3	Iteration assessments and change control, Elaboration	Boonnisa T.	28	-	-	-	28	9,800
1.4	Iteration assessments and change control, Release 1.0 Construction	Boonnisa T.	64	-	-	-	64	22,400

WBS ID	Work package (* three-point, Table 4.2)	Accountable	Boonnisa T. (PM)	Sirapatr C. (SA)	Chawanakorn S. (RS)	Watayut A. (QA)	Hours	Labor (THB)
1.5	Iteration assessments and change control, Release 1.0 Transition	Boonnisa T.	12	-	-	-	12	4,200
1.6	Iteration assessments and change control, Release 2.0	Boonnisa T.	46	-	-	-	46	16,100
1.7	Iteration assessments and change control, Release 2.1	Boonnisa T.	28	-	-	-	28	9,800
1.8	Pilot KPI reports and reviews	Boonnisa T.	12	-	-	8	20	6,600
1.9	Pilot review, expansion decision and close-out	Boonnisa T.	8	4	-	4	16	5,400
2	**Inception (I1, I2)**							
2.1	Stakeholder requests and problem analysis	Sirapatr C.	4	16	8	-	28	9,400
2.2	Vision and Glossary	Sirapatr C.	-	20	12	-	32	10,600
2.3	Use-case model and release plan	Chawanakorn S.	-	4	12	4	20	6,200
2.4	Review records and LCO package	Watayut A.	4	-	-	12	16	5,000
3	**Elaboration (E1, E2)**							
3.1	Use-case specifications	Chawanakorn S.	-	16	32	4	52	16,400
3.2	Software Requirements Specification and Supplementary Specification	Chawanakorn S.	4	12	28	8	52	16,400
3.3	Analysis model and Software Architecture Document	Sirapatr C.	4	48	8	-	60	20,600
3.4	User-interface prototype	Chawanakorn S.	-	4	24	4	32	9,800
3.5	Payment gateway and LINE sandbox	Sirapatr C.	8	12	4	-	24	8,200
3.6	Test plan and test cases	Watayut A.	4	-	-	32	36	11,000
3.7	Requirements reviews, SRS baseline and LCA package	Watayut A.	8	4	-	16	28	9,000
3.8	Scope baseline, estimates, schedule and cost baseline	Boonnisa T.	8	-	-	-	8	2,800
4	**Release 1.0 Core: Construction (C1, C2)**							
4.1	Concert rounds, zones and packages	Chawanakorn S.	8	16	32	8	64	20,400
4.2	Table map, selection and hold	Sirapatr C.	8	32	36	16	92	29,600

WBS ID	Work package (* three-point, Table 4.2)	Accountable	Boonnisa T. (PM)	Sirapatr C. (SA)	Chawanakorn S. (RS)	Watayut A. (QA)	Hours	Labor (THB)
4.3	Gateway payment and callback *	Sirapatr C.	8	40	32	24	104	33,600
4.4	LINE sign-in, staff accounts and role access	Chawanakorn S.	-	16	16	8	40	12,800
4.5	C1 integration and iteration test	Watayut A.	8	8	8	24	48	15,200
4.6	QR e-ticket and transactional messages	Chawanakorn S.	-	16	32	8	56	17,600
4.7	Door check-in and name lookup	Chawanakorn S.	8	16	32	16	72	22,800
4.8	Live booking view and exception handling	Sirapatr C.	8	32	32	16	88	28,400
4.9	Audit trail, backup, consent and push budget	Chawanakorn S.	8	16	20	8	52	16,800
4.10	Load test, system test and IOC package *	Watayut A.	8	32	16	64	120	38,000
5	**Release 1.0 Core: Transition (T1)**							
5.1	Acceptance tests of Release 1.0	Watayut A.	8	8	12	28	56	17,600
5.2	Training and user guide	Chawanakorn S.	8	-	16	4	28	8,800
5.3	Production set-up and data migration	Sirapatr C.	-	24	10	8	42	13,800
5.4	Pilot concert nights and go-live of Release 1.0	Boonnisa T.	12	8	12	20	52	16,600
6	**Release 2.0 Growth (C3, T2)**							
6.1	Scheduled reminders *	Chawanakorn S.	8	32	40	20	100	32,000
6.2	Grace extension and waitlist	Sirapatr C.	8	40	48	28	124	39,600
6.3	Booking transfer	Chawanakorn S.	-	16	24	12	52	16,400
6.4	Reports and Excel export	Chawanakorn S.	8	24	32	16	80	25,600
6.5	C3 integration and system test	Watayut A.	12	32	24	44	112	35,800
6.6	Acceptance tests and training on Release 2.0	Watayut A.	8	14	24	30	76	23,900
6.7	Release 2.0 deployment and first concert nights	Boonnisa T.	12	16	16	20	64	20,600
7	**Pilot operation (P1)**							

WBS ID	Work package (* three-point, Table 4.2)	Accountable	Boonnisa T. (PM)	Sirapatr C. (SA)	Chawanakorn S. (RS)	Watayut A. (QA)	Hours	Labor (THB)
7.1	Concert-night support and maintenance (operating expenditure, outside the 2,632 h)	Watayut A.	-	-	48	48	96	28,800
8	**Release 2.1 Product (C4, T3)**							
8.1	Multi-tenant data model and tenant isolation *	Sirapatr C.	-	40	40	16	96	30,800
8.2	Plans, limits and tier enforcement	Chawanakorn S.	-	20	40	20	80	25,000
8.3	Subscription billing and company set-up	Chawanakorn S.	8	24	40	24	96	30,400
8.4	Self-service onboarding and service administration	Chawanakorn S.	-	20	40	20	80	25,000
8.5	Security, load and integration test of Release 2.1	Watayut A.	8	12	20	28	68	21,400
8.6	Migration of La Loy and release of the Free-plan sign-up	Sirapatr C.	4	12	20	20	56	17,600
Total development effort			**434**	**706**	**842**	**650**	**2632**	**846,600**
Rate (THB/h)			350	350	300	300		
Labor cost (THB)			**151,900**	**247,100**	**252,600**	**195,000**		**846,600**

4.3 Three-point Estimates of the Risky Work Packages

The four work packages that carry the technical risks of Milestone 1 (RISK-03 gateway onboarding, RISK-05 burst load, the LINE push quota of RISK-07, the multi-tenant architecture of Release 2.1) were estimated three-point by the engineers who will build them, with the PERT weighted average (O + 4M + P) / 6 (Table 4.2). The allocated hours are within 2.5 % of the PERT values; the gap between the most likely and the pessimistic values is why these packages are named in the contingency reserve (Section 6.2).

Table 4.2 Three-point estimates and PERT weighted average

WBS ID	Work package	Optimistic O (h)	Most likely M (h)	Pessimistic P (h)	PERT (O + 4M + P) / 6	Allocated (h)	Allocated vs PERT
4.3	Gateway payment and callback	80	100	150	105.0	104	-1.0 %
4.10	Load test, system test and IOC package	90	115	180	121.7	120	-1.4 %

WBS ID	Work package	Optimistic O (h)	Most likely M (h)	Pessimistic P (h)	PERT (O + 4M + P) / 6	Allocated (h)	Allocated vs PERT
6.1	Scheduled reminders	80	96	150	102.3	100	-2.3 %
8.1	Multi-tenant data model and tenant isolation	72	92	150	98.3	96	-2.4 %

4.4 Comparison with the Top-down Estimate of Milestone 1

Table 4.3 compares the bottom-up hours per phase and engineer with the top-down estimate of Milestone 1 (Table 3.4 there). The totals per engineer and per year are identical by the Review Board's decision; per phase the bottom-up plan differs in two places: the monthly pilot KPI reports and the close-out (1.8, 1.9, 36 hours) are drawn from the Release 2.1 pool, which Milestone 1 had no row for, and 3.5 hours of the Project Manager's control work move from C3 to T2 because work package 1.6 spans both iterations. The percent difference is below 8 % for every phase that has hours in both estimates and 0 % in total, which confirms the Milestone 1 figures as the planning baseline.

Table 4.3 Bottom-up versus top-down effort by phase

Phase	Boonnisa T.	Sirapatr C.	Chawanakorn S.	Watayut A.	Bottom-up (h)	Milestone 1 top-down (h)	Difference (h)	Percent difference
Inception (I1, I2)	32	40	32	24	128	128	+0	0.0 %
Elaboration (E1, E2)	64	96	96	64	320	320	+0	0.0 %
Construction Release 1.0 (C1, C2)	128	224	256	192	800	800	+0	0.0 %
Transition Release 1.0 (T1)	40	40	50	60	190	190	+0	0.0 %
Construction Release 2.0 (C3)	68.4706	144	168	120	500.471	504	-3.52941	0.7 %
Transition Release 2.0 (T2)	33.5294	30	40	50	153.529	150	+3.52941	2.3 %
Construction Release 2.1 (C4)	40	116	180	108	444	480	-36	7.8 %
Transition Release 2.1 (T3)	8	12	20	20	60	60	+0	0.0 %
Pilot operation and close-out (P1)	20	4	0	12	36	0	+36	200.0 %
Total	**434**	**706**	**842**	**650**	**2632**	**2632**	**+0**	**0.0 %**

5. Human and Physical Resource Assignment and Rates

5.1 Resource Rates

Table 5.1 lists the human resources with their roles, cost rates and availability per phase (the hours a week of Table 3.4 of Milestone 1, which are the maximum units of each engineer in the schedule), and the physical resources (equipment, services, legal set-up) with their cost basis and the work packages that use them. Rates are per hour and identical to Milestone 1; items paid by the restaurant are listed so that they are not confused with project costs.

Table 5.1 Resource rate table (human and physical resources)

Resource	Role / use	Type	Rate or cost basis	Availability (hours a week per phase) / when
Boonnisa Tangjitpermkwamdee	Project Manager, Change Control Manager	Work (human)	350 THB/h	Inception 8; Elaboration 8; C 1.0 8; T 1.0 8; C 2.0 6; T 2.0 6; C 2.1 2.5; T 2.1 2 h/week
Sirapatr Chalermwattanatrai	System Analyst, Software Architect	Work (human)	350 THB/h	Inception 10; Elaboration 12; C 1.0 14; T 1.0 8; C 2.0 12; T 2.0 6; C 2.1 5; T 2.1 3 h/week
Chawanakorn Sanannam	Requirements Specifier, User-Interface Designer, Designer / Implementer	Work (human)	300 THB/h	Inception 8; Elaboration 12; C 1.0 16; T 1.0 10; C 2.0 14; T 2.0 8; C 2.1 7.5; T 2.1 5 h/week
Watayut Aiamanan	Quality Assurance, Test Designer / Tester, Implementer (Construction)	Work (human)	300 THB/h	Inception 6; Elaboration 8; C 1.0 12; T 1.0 12; C 2.0 10; T 2.0 10; C 2.1 5; T 2.1 5 h/week
Android phone and 10-inch tablet	LIFF and check-in testing (4.2 to 4.10, 5.1, 6.5, 8.5)	Equipment (CAPEX)	15,000 THB, one set	C1
Development and staging hosting	Build and test environments (3.3 to 8.5)	Service	300 THB a month, 12 months of Year 0	I1 to T2

Resource	Role / use	Type	Rate or cost basis	Availability (hours a week per phase) / when
Production hosting	Pilot restaurant's live system (5.3 onwards)	Service	700 THB a month: 5 months of Year 0, 12 of Year 1	T1 to P1
Domain name and TLS certificate	Production set-up (5.3)	Service	1,200 THB a year	T1
Payment gateway sandbox, LINE Developers, GitHub, Figma, PlantUML	3.5 onwards	Service	free tiers, 0 THB	E2 onwards
Company registration, terms of service, privacy policy, data-processing agreement	Subscription business (8.3)	Legal (Year 1)	15,000 THB, once	C4
Pilot restaurant's check-in tablet, LINE Official Account plan, gateway merchant account	Operation at La Loy	Customer-paid	paid by La Loy (CON-03, CON-07); about 9,000 THB for the tablet	T1 onwards

5.2 Assignment of Resources to Work Packages

Human resources are assigned to the work packages in Table 4.1 (accountable engineer and hours of every engineer) and Table 3.1 (units per activity); physical resources are assigned in Table 5.1. Table 5.2 summarizes the assignment per branch as a responsibility assignment matrix in RACI form: A/R the engineer accountable for the listed work packages (who also works on them), R the other engineers assigned with hours, C the restaurant-side stakeholders and external parties consulted, I the parties kept informed through the iteration assessments. The matrix keeps the rules of Milestone 1: one owner per artifact, and the reviewer (Quality Assurance) is never the author of what it reviews.

Table 5.2 Responsibility assignment matrix (RACI) by WBS branch

WBS	Branch	Boonnisa T. (Project Manager)	Sirapatr C. (System Analyst)	Chawanakorn S. (Requirements Specifier)	Watayut A. (Quality Assurance)	Restaurant Owner	Manager / Front Staff	Other consulted
1	Project Management	A/R (1.1, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9)	R	I	A/R (1.2)	C	C	Sponsor (partners)

WBS	Branch	Boonnisa T. (Project Manager)	Sirapatr C. (System Analyst)	Chawanakorn S. (Requirements Specifier)	Watayut A. (Quality Assurance)	Restaurant Owner	Manager / Front Staff	Other consulted
2	Inception (I1, I2)	R	A/R (2.1, 2.2)	A/R (2.3)	A/R (2.4)	C	C	-
3	Elaboration (E1, E2)	A/R (3.8)	A/R (3.3, 3.5)	A/R (3.1, 3.2, 3.4)	A/R (3.6, 3.7)	C	C	Customers, Gateway provider, LINE Platform, Sponsor (partners)
4	Release 1.0 Core: Construction (C1, C2)	R	A/R (4.2, 4.3, 4.8)	A/R (4.1, 4.4, 4.6, 4.7, 4.9)	A/R (4.5, 4.10)	I	C	Gateway provider
5	Release 1.0 Core: Transition (T1)	A/R (5.4)	A/R (5.3)	A/R (5.2)	A/R (5.1)	C	C	Gateway provider
6	Release 2.0 Growth (C3, T2)	A/R (6.7)	A/R (6.2)	A/R (6.1, 6.3, 6.4)	A/R (6.5, 6.6)	C	C	-
7	Pilot operation (P1)	I	I	R	A/R (7.1)	I	C	-
8	Release 2.1 Product (C4, T3)	R	A/R (8.1, 8.6)	A/R (8.2, 8.3, 8.4)	A/R (8.5)	C	C	Legal counsel, Sponsor (partners)

5.3 Resource Loading and Leveling

Table 5.3 compares, per phase and engineer, the planned average hours a week with the availability of Milestone 1, and shows the peak week when the hours of each work package are spread evenly over its duration. The averages match the availability by construction. Peaks above it occur where several packages of one engineer overlap inside a Construction iteration; two were flattened by starting the test plan (3.6) and the Release 2.1 test (8.5) earlier, and the rest are leveled within the iteration by the engineer working the overlapping packages one after the other inside their free slack (every package but the last of an iteration has slack up to the iteration's test package), because the software house commits each engineer to the phase total of hours, not to a weekly cap. No leveling moves a milestone; if other client work makes a peak impossible, the response of RISK-02 applies and the priority-2 features of the iteration move to the next one.

Table 5.3 Resource loading per phase: planned average and peak week against availability

Phase	Weeks	Boonnisa T.: hours a week (planned avg, peak / available)	Sirapatr C.: hours a week (planned avg, peak / available)	Chawanakorn S.: hours a week (planned avg, peak / available)	Watayut A.: hours a week (planned avg, peak / available)
Inception (I1, I2)	4	8.0 avg, 7 peak / 8	10.0 avg, 11 peak / 10	8.0 avg, 14 peak / 8	6.0 avg, 11 peak / 6
Elaboration (E1, E2)	8	8.0 avg, 14 peak / 8	12.0 avg, 18 peak / 12	12.0 avg, 19 peak / 12	8.0 avg, 16 peak / 8
Construction Release 1.0 (C1, C2)	16	8.0 avg, 11 peak / 8	14.0 avg, 24 peak / 14	16.0 avg, 27 peak / 16	12.0 avg, 28 peak / 12
Transition Release 1.0 (T1)	5	8.0 avg, 10 peak / 8	8.0 avg, 16 peak / 8	10.0 avg, 14 peak / 10	12.0 avg, 16 peak / 12
Construction Release 2.0 (C3)	12	5.7 avg, 9 peak / 6	12.0 avg, 18 peak / 12	14.0 avg, 24 peak / 14	10.0 avg, 19 peak / 10
Transition Release 2.0 (T2)	5	6.7 avg, 9 peak / 6	6.0 avg, 10 peak / 6	8.0 avg, 13 peak / 8	10.0 avg, 17 peak / 10
Construction Release 2.1 (C4)	24	1.7 avg, 3 peak / 2.5	4.8 avg, 8 peak / 5	7.5 avg, 13 peak / 7.5	4.5 avg, 9 peak / 5
Transition Release 2.1 (T3)	4	2.0 avg, 2 peak / 2	3.0 avg, 3 peak / 3	5.0 avg, 5 peak / 5	5.0 avg, 5 peak / 5

6. Cost Estimate

6.1 Cost by WBS Item

The cost baseline is built up from the work packages, as the PMBOK budget build-up prescribes: the labor of each work package is its hours times the rate of each assigned engineer (Table 4.1), the other costs are the physical resources of Table 5.1 attached to the work packages that use them, and the contingency reserve for the identified risks is added to form the cost baseline (Table 6.1). The totals equal the budget of the charter and the cost tables of Milestone 1 (Tables 2.20 and 3.7 there): labor 846,600 THB, other costs 75,500 THB (test devices 15,000, hosting and domain 8,300, legal set-up 15,000, pilot hosting 8,400, pilot support 28,800), contingency 88,490 THB.

Table 6.1 Cost estimate by WBS item, contingency reserve, cost baseline and project budget

WBS	Item	Hours	Labor basis (hours x THB/h)	Labor (THB)	Other costs (THB)	Subtotal (THB)
1	Project Management	246	Boonnisa T. 222 h x 350; Sirapatr C. 4 h x 350; Watayut A. 20 h x 300	85,100	-	85,100
2	Inception (I1, I2)	96	Boonnisa T. 8 h x 350; Sirapatr C. 40 h x 350; Chawanakorn S. 32 h x 300; Watayut A. 16 h x 300	31,200	Development and staging hosting, August 2026 300	31,500
3	Elaboration (E1, E2)	292	Boonnisa T. 36 h x 350; Sirapatr C. 96 h x 350; Chawanakorn S. 96 h x 300; Watayut A. 64 h x 300	94,200	Development and staging hosting, September to October 2026 600	94,800
4	Release 1.0 Core: Construction (C1, C2)	736	Boonnisa T. 64 h x 350; Sirapatr C. 224 h x 350; Chawanakorn S. 256 h x 300; Watayut A. 192 h x 300	235,200	Test devices: Android phone and 10-inch tablet 15,000; Development and staging hosting, November 2026 to February 2027 1,200	251,400
5	Release 1.0 Core: Transition (T1)	178	Boonnisa T. 28 h x 350; Sirapatr C. 40 h x 350; Chawanakorn S. 50 h x 300; Watayut A. 60 h x 300	56,800	Domain name and TLS certificate 1,200; Development and staging hosting, March 2027 300; Production hosting of the pilot restaurant, March 2027 700	59,000
6	Release 2.0 Growth (C3, T2)	608	Boonnisa T. 56 h x 350; Sirapatr C. 174 h x 350; Chawanakorn S. 208 h x 300; Watayut A. 170 h x 300	193,900	Development and staging hosting, April to July 2027 1,200; Production hosting of the pilot restaurant, April to July 2027 2,800	197,900
7	Pilot operation (P1)	0		0	Production hosting during the pilot year 8,400; support labor 96 h x 300 = 28,800	37,200

WBS	Item	Hours	Labor basis (hours x THB/h)	Labor (THB)	Other costs (THB)	Subtotal (THB)
8	Release 2.1 Product (C4, T3)	476	Boonnisa T. 20 h x 350; Sirapatr C. 128 h x 350; Chawanakorn S. 200 h x 300; Watayut A. 128 h x 300	150,200	Company registration, terms of service, privacy policy and data-processing agreement 15,000	165,200
	Work-package estimates	**2632**		**846,600**	**75,500**	**922,100**
	Contingency reserve, Year 0 (RISK-02, RISK-03, RISK-05, RISK-08, RISK-10)		10 % of 697,900 THB			69,790
	Contingency reserve, Year 1 (RISK-04, RISK-07, RISK-12)		10 % of 187,000 THB (labor of C4 and T3 and the legal set-up)			18,700
	Cost baseline					**1,010,590**
	Management reserve		none held by the sponsor for this project (unknown unknowns escalate to the partners' meeting)			0
	Project budget (charter)					**1,010,590**
	Less: pilot implementation fee of La Loy (revenue inside the project)					(150,000)
	Net investment of the software house					**860,590**

6.2 Contingency Reserve and Management Reserve

The contingency reserve covers identified risks (known unknowns) and is part of the cost baseline. Year 0 holds 69,790 THB, 10 % of the Year 0 cost, for the risks that can lengthen the build: RISK-02 (part-time engineers delayed by other work, about 2 weeks of an iteration, 35,000 THB), RISK-03 (gateway onboarding or LINE settings late, a re-run of the gateway package, 20,000 THB), RISK-05 (burst load above design, an extra load-test and tuning cycle, 10,000 THB), RISK-08 and RISK-10 (late business rules and change requests after the SRS baseline, rework in C3, 5,000 THB). Year 1 holds 18,700 THB, 10 % of the productization labor and the legal set-up, for RISK-04 (support and productization competing for the same engineers), RISK-07 (push messages or hosting above the operating-cost ceiling) and RISK-12 (slower adoption, which may require a second pilot venue before the gate). The reserve is released by the Project Manager at an iteration assessment against a recorded risk event, never for new features. The sponsor holds no management reserve for this project: unknown unknowns are escalated to the partners' meeting of the software house, which decides on

additional funds or a scope change through the Change Control Board; the project budget therefore equals the cost baseline, 1,010,590 THB.

6.3 Time-phased Cost Baseline

Table 6.2 phases the cost baseline over the iterations: the labor of each work package is spread evenly over its working days, the other costs fall in the iteration that uses them, and the contingency reserve of each year is distributed in proportion to the cost of the iterations of that year. The cumulative column is the S-curve against which cost performance is measured at every iteration assessment; the cumulative at the end of T2 is the 767,690 THB of the Year 0 budget.

Table 6.2 Time-phased cost baseline by iteration

Iteration	Name	Period	Labor (THB)	Other costs (THB)	Contingency (THB)	Subtotal (THB)	Cumulative (THB)
I1	Inception I1	3 Aug 2026 to 24 Aug 2026	23,070	300	2,337	25,707	25,707
I2	Inception I2	25 Aug 2026 to 2 Sep 2026	18,930	0	1,893	20,823	46,530
E1	Elaboration E1	3 Sep 2026 to 16 Sep 2026	22,776	600	2,338	25,714	72,244
E2	Elaboration E2	17 Sep 2026 to 23 Oct 2026	81,224	0	8,122	89,346	161,590
C1	Construction C1	26 Oct 2026 to 18 Dec 2026	122,800	16,200	13,900	152,900	314,490
C2	Construction C2	4 Jan 2027 to 26 Feb 2027	134,800	0	13,480	148,280	462,770
T1	Transition T1	1 Mar 2027 to 2 Apr 2027	61,000	2,200	6,320	69,520	532,290
C3	Construction C3	5 Apr 2027 to 25 Jun 2027	160,765	0	16,076	176,841	709,131
T2	Transition T2	28 Jun 2027 to 30 Jul 2027	49,235	4,000	5,324	58,559	767,690
P1	Pilot year P1	2 Aug 2027 to 28 Jul 2028	6,600	37,200	3,653	47,453	815,143
C4	Construction C4	2 Aug 2027 to 14 Jan 2028	141,000	15,000	13,012	169,012	984,155
T3	Transition T3	17 Jan 2028 to 11 Feb 2028	19,000	0	1,585	20,585	1,004,740

Iteration	Name	Period	Labor (THB)	Other costs (THB)	Contingency (THB)	Subtotal (THB)	Cumulative (THB)
CL	Close-out	24 Jul 2028 to 28 Jul 2028	5,400	0	450	5,850	1,010,590
	Total		**846,600**	**75,500**	**88,490**	**1,010,590**	

6.4 Reconciliation and Cost-control Rules

The bottom-up cost baseline reconciles with Milestone 1 without variance: 846,600 THB of labor for 2,632 hours, 75,500 THB of other costs, 88,490 THB of contingency, 1,010,590 THB in total, of which 767,690 THB in Year 0 and 242,900 THB in Year 1; less the pilot implementation fee of 150,000 THB, the net investment of the software house is 860,590 THB, as in the charter. The cost-control rules of the project are:

- The Project Manager tracks actual hours against the 2,632-hour plan and actual cost against the time-phased baseline at every iteration assessment; the control threshold is ±10 % of the cumulative baseline, above which a change request or a re-plan is raised.
- Any change that affects scope, budget or milestone dates is assessed for impact and decided by the Change Control Board; the baseline is re-planned, not silently extended (RISK-10).
- The contingency reserve is held for identified risks and released against a recorded risk event; it is not planned feature-development effort.
- The budget excludes the costs that Milestone 1 assigns to the restaurant: its LINE plan, the gateway's merchant fees and the check-in tablet.
- Figures are kept in 2026 THB, rounded to the nearest 10 THB in the tables and to the hour in the effort plan.

7. Schedule, Gantt Chart and Critical Path

7.1 Gantt Chart

Figure 7.1 is the Gantt chart of the schedule baseline: the 49 activities of Table 3.1 with their assigned hours, grouped by WBS branch over the iteration bands of the tailored RUP, with the seven milestones. The schedule keeps every iteration and milestone date of Milestone 1; the activities inside an iteration overlap by start-to-start links so that the engineers' part-time hours fit the iteration, and the test and integration package of every Construction iteration ends on the iteration's last day.

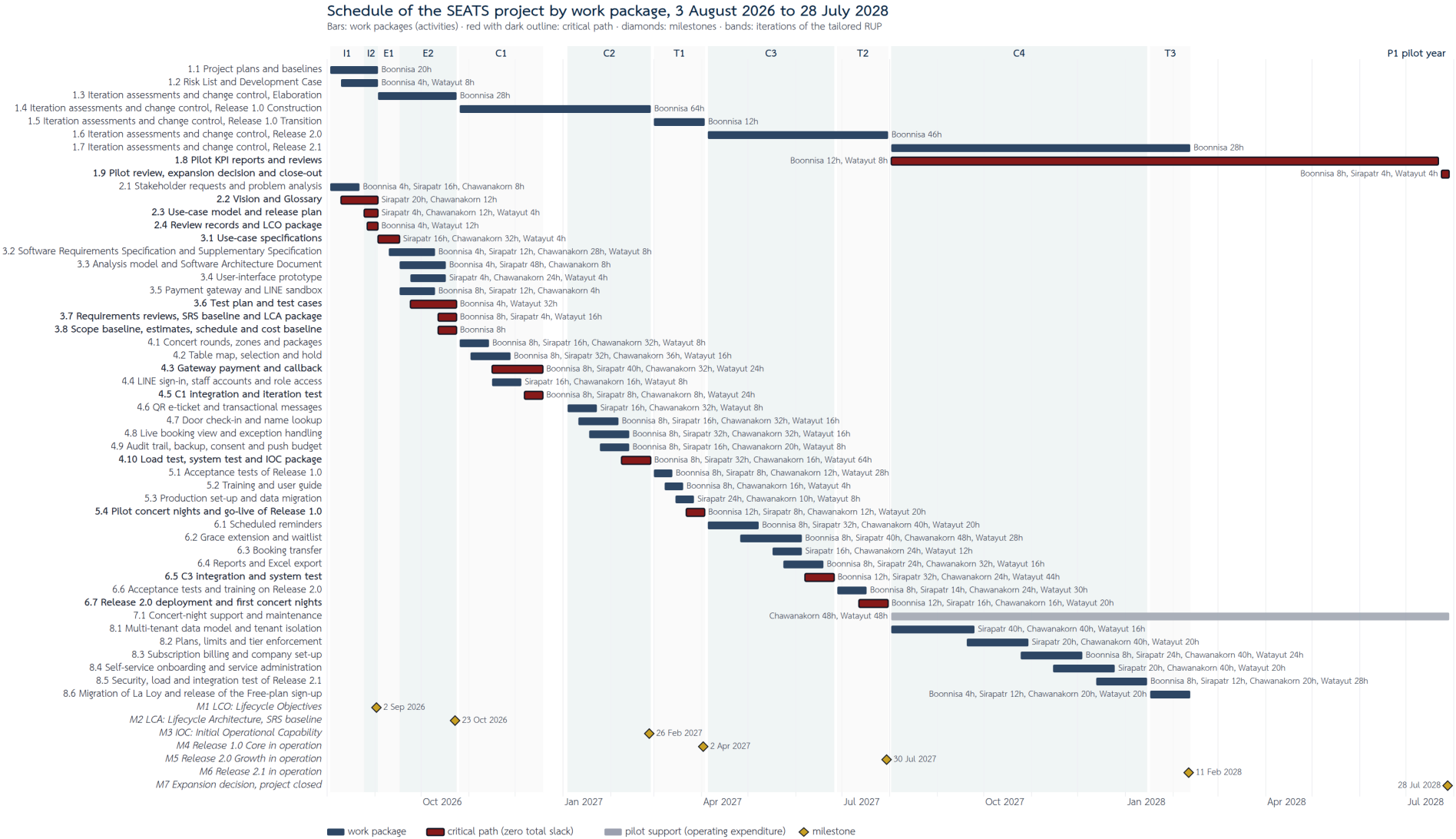


Figure 7.1 Gantt chart of the SEATS project by work package, with iterations, milestones and critical path

7.2 Critical Path

The critical path is calculated on the iteration network with the forward and backward pass in working days (Table 7.1): ES and EF from the predecessors, LS and LF from the project finish, total slack TS = LS − ES = LF − EF. The iterations are linked finish-to-start as the Development Case prescribes; the pilot year P1 and the productization branch C4, T3 both start after Release 2.0 is in operation, and the close-out follows T3 and finishes together with the pilot year (finish-to-finish), because the expansion decision needs both the pilot KPIs and the productized service. The project is 510 working days long. The zero-slack path is I1, I2, E1, E2, C1, C2, T1, C3, T2, P1 and the close-out: every iteration up to Release 2.0 is critical, because each starts when the previous one ends, and after Release 2.0 the pilot year drives the finish date, since its 260 working days of KPI measurement are longer than the 140 working days of C4 and T3. The productization branch has 115 working days of total slack, so a delay of Release 2.1 by up to 23 weeks would not move the expansion decision; a delay of any iteration up to T2, or a shorter pilot year, would. Within the critical iterations the critical work packages are those that end on the iteration's last day and feed the next iteration, marked red in Figure 7.1: the LCO package (2.4), the SRS baseline and LCA package (3.7), the gateway payment (4.3) and the C1 integration test (4.5), the load and system test before IOC (4.10), the go-live of Release 1.0 (5.4), the C3 integration test (6.5), the deployment of Release 2.0 (6.7), the pilot KPI reports (1.8) and the close-out (1.9). This is a planning analysis; if an iteration threatens its milestone, the responses are, in order, the re-planning of priority-2 features (RISK-02), fast-tracking the next iteration's design work in parallel with the current iteration's test package, and crashing by raising the engineers' hours a week within the contingency reserve.

Table 7.1 Critical path method on the iteration network (working days from 3 August 2026)

ID	Iteration	Duration (wd)	Predecessors	ES	EF	LS	LF	Total slack (wd)	Critical	Early start	Early finish
I1	Inception I1	16	—	0	16	0	16	0	Yes	3 Aug 2026	24 Aug 2026
I2	Inception I2	7	I1	16	23	16	23	0	Yes	25 Aug 2026	2 Sep 2026
E1	Elaboration E1	10	I2	23	33	23	33	0	Yes	3 Sep 2026	16 Sep 2026
E2	Elaboration E2	27	E1	33	60	33	60	0	Yes	17 Sep 2026	23 Oct 2026
C1	Construction C1	40	E2	60	100	60	100	0	Yes	26 Oct 2026	18 Dec 2026

ID	Iteration	Duration (wd)	Predecessors	ES	EF	LS	LF	Total slack (wd)	Critical	Early start	Early finish
C2	Construction C2	40	C1	100	140	100	140	0	Yes	4 Jan 2027	26 Feb 2027
T1	Transition T1	25	C2	140	165	140	165	0	Yes	1 Mar 2027	2 Apr 2027
C3	Construction C3	60	T1	165	225	165	225	0	Yes	5 Apr 2027	25 Jun 2027
T2	Transition T2	25	C3	225	250	225	250	0	Yes	28 Jun 2027	30 Jul 2027
P1	Pilot year P1	260	T2	250	510	250	510	0	Yes	2 Aug 2027	28 Jul 2028
C4	Construction C4	120	T2	250	370	365	485	115	No	2 Aug 2027	14 Jan 2028
T3	Transition T3	20	C4	370	390	485	505	115	No	17 Jan 2028	11 Feb 2028
CL	Close-out	5	T3, P1 (FF)	505	510	505	510	0	Yes	24 Jul 2028	28 Jul 2028

7.3 Milestone Control Points

Table 7.2 Milestones and their acceptance decisions

ID	Milestone	Date	After	Acceptance / decision
M1	LCO: Lifecycle Objectives	2 Sep 2026	2.4	Vision approved by the Owner; Business Case, charter, Development Case and plans accepted; go for Elaboration
M2	LCA: Lifecycle Architecture, SRS baseline	23 Oct 2026	3.7	SRS signed off by the Owner; architecture, prototype and test plan accepted; gateway and LINE prerequisites verified; financial analysis re-run; go for Construction
M3	IOC: Initial Operational Capability	26 Feb 2027	4.10	Release 1.0 passes the system and load tests (5 transactions a second); go for the pilot on concert nights
M4	Release 1.0 Core in operation	2 Apr 2027	5.4	Acceptance tests of the two main use cases passed with the Manager and Front Staff; data migrated; Release 1.0 declared in operation
M5	Release 2.0 Growth in operation	30 Jul 2027	6.7	Acceptance tests of the Release 2.0 features passed; pilot year and KPI measurement begin
M6	Release 2.1 in operation	11 Feb 2028	8.6	La Loy migrated to the productized service; Free-plan sign-up released; hand-over to the support routine
M7	Expansion decision, project closed	28 Jul 2028	1.9	Pilot review against the seven success criteria of the charter; expansion decision of the software house; lessons learned; sign-off